24. Peripheral DMA Controller (PDC)

24.1 Description

The Peripheral DMA Controller (PDC) transfers data between on-chip serial peripherals and the on- and/or off-chip memories. The link between the PDC and a serial peripheral is operated by the AHB to ABP bridge.

The PDC contains 22 channels. The full-duplex peripherals feature 21 mono directional channels used in pairs (transmit only or receive only). The half-duplex peripherals feature 1 bi-directional channels.

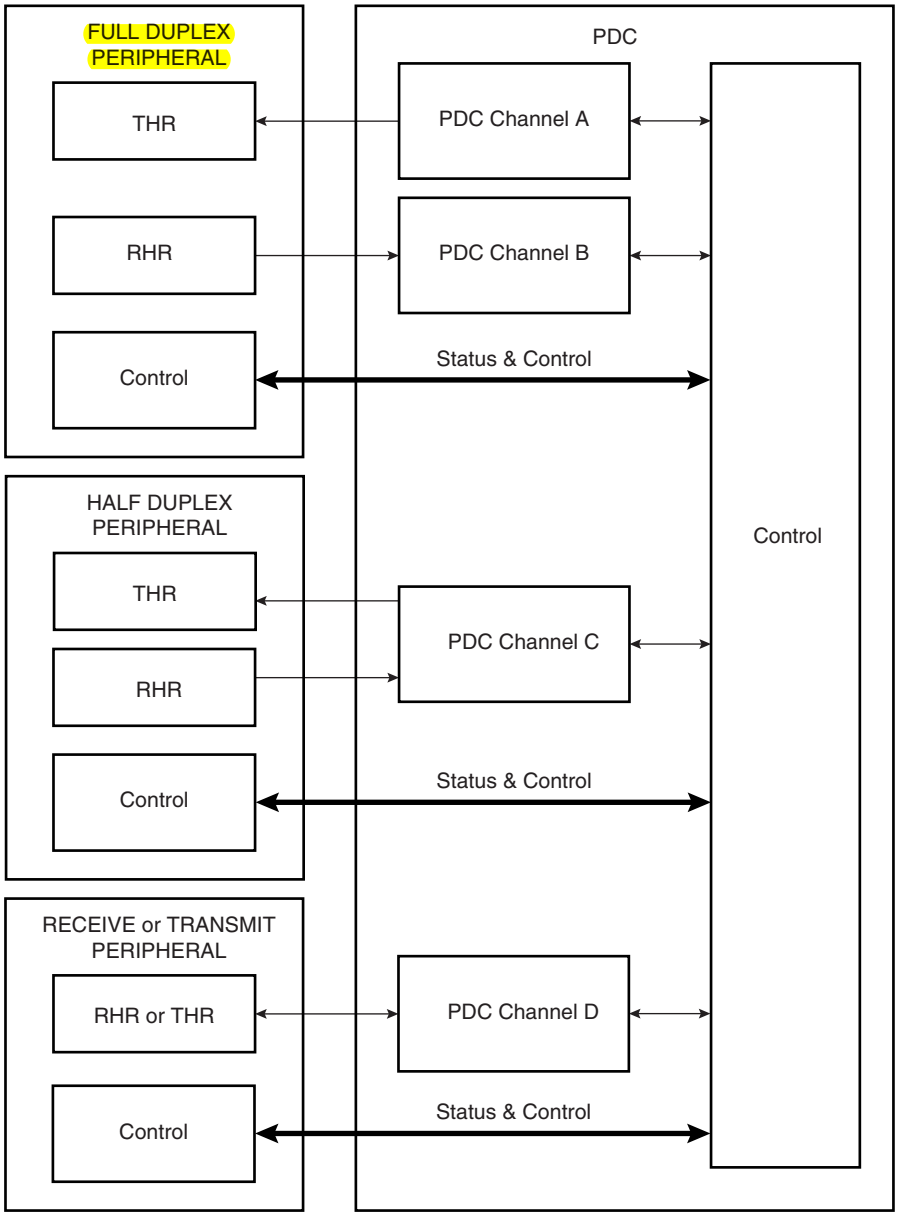
The user interface of each PDC channel is integrated into the user interface of the peripheral it serves. The user interface of mono directional channels (receive only or transmit only), contains two 32-bit memory pointers and two 16-bit counters, one set (pointer, counter) for current transfer and one set (pointer, counter) for next transfer. The bi-directional channel user interface contains four 32-bit memory pointers and four 16-bit counters. Each set (pointer, counter) is used by current transmit, next transmit, current receive and next receive.

Using the PDC removes processor overhead by reducing its intervention during the transfer. This significantly reduces the number of clock cycles required for a data transfer, which improves microcontroller performance.

To launch a transfer, the peripheral triggers its associated PDC channels by using transmit and receive signals. When the programmed data is transferred, an end of transfer interrupt is generated by the peripheral itself.

24.2 Block Diagram

Figure 24-1. Block Diagram



24.3 Functional Description

24.3.1 Configuration

The PDC channel user interface enables the user to configure and control data transfers for each channel. The user interface of each PDC channel is integrated into the associated peripheral user interface.

The user interface of a serial peripheral, whether it is full or half duplex, contains four 32-bit pointers (RPR, RNPR, TPR, TNPR) and four 16-bit counter registers (RCR, RNCR, TCR, TNCR). However, the transmit and receive parts of each type are programmed differently: the

transmit and receive parts of a full duplex peripheral can be programmed at the same time, whereas only one part (transmit or receive) of a half duplex peripheral can be programmed at a time.

32-bit pointers define the access location in memory for current and next transfer, whether it is for read (transmit) or write (receive). 16-bit counters define the size of current and next transfers. It is possible, at any moment, to read the number of transfers left for each channel.

The PDC has dedicated status registers which indicate if the transfer is enabled or disabled for each channel. The status for each channel is located in the associated peripheral status register. Transfers can be enabled and/or disabled by setting TXTEN/TXTDIS and RXTEN/RXTDIS in the peripheral's Transfer Control Register.

At the end of a transfer, the PDC channel sends status flags to its associated peripheral. These flags are visible in the peripheral status register (ENDRX, ENDTX, RXBUFF, and TXBUFE). Refer to Section 24.3.3 and to the associated peripheral user interface.

24.3.2 Memory Pointers

Each full duplex peripheral is connected to the PDC by a receive channel and a transmit channel. Both channels have 32-bit memory pointers that point respectively to a receive area and to a transmit area in on- and/or off-chip memory.

Each half duplex peripheral is connected to the PDC by a bidirectional channel. This channel has two 32-bit memory pointers, one for current transfer and the other for next transfer. These pointers point to transmit or receive data depending on the operating mode of the peripheral.

Depending on the type of transfer (byte, half-word or word), the memory pointer is incremented respectively by 1, 2 or 4 bytes.

If a memory pointer address changes in the middle of a transfer, the PDC channel continues operating using the new address.

24.3.3 Transfer Counters

Each channel has two 16-bit counters, one for current transfer and the other one for next transfer. These counters define the size of data to be transferred by the channel. The current transfer counter is decremented first as the data addressed by current memory pointer starts to be transferred. When the current transfer counter reaches zero, the channel checks its next transfer counter. If the value of next counter is zero, the channel stops transferring data and sets the appropriate flag. But if the next counter value is greater then zero, the values of the next pointer/next counter are copied into the current pointer/current counter and the channel resumes the transfer whereas next pointer/next counter get zero/zero as values. At the end of this transfer the PDC channel sets the appropriate flags in the Peripheral Status Register.

The following list gives an overview of how status register flags behave depending on the counters' values:

- ENDRX flag is set when the PERIPH_RCR register reaches zero.
- RXBUFF flag is set when both PERIPH_RCR and PERIPH_RNCR reach zero.
- ENDTX flag is set when the PERIPH_TCR register reaches zero.
- TXBUFE flag is set when both PERIPH_TCR and PERIPH_TNCR reach zero.

These status flags are described in the Peripheral Status Register.

24.3.4 Data Transfers

The serial peripheral triggers its associated PDC channels' transfers using transmit enable (TXEN) and receive enable (RXEN) flags in the transfer control register integrated in the peripheral's user interface.

When the peripheral receives an external data, it sends a Receive Ready signal to its PDC receive channel which then requests access to the Matrix. When access is granted, the PDC receive channel starts reading the peripheral Receive Holding Register (RHR). The read data are stored in an internal buffer and then written to memory.

When the peripheral is about to send data, it sends a Transmit Ready to its PDC transmit channel which then requests access to the Matrix. When access is granted, the PDC transmit channel reads data from memory and puts them to Transmit Holding Register (THR) of its associated peripheral. The same peripheral sends data according to its mechanism.

24.3.5 PDC Flags and Peripheral Status Register

Each peripheral connected to the PDC sends out receive ready and transmit ready flags and the PDC sends back flags to the peripheral. All these flags are only visible in the Peripheral Status Register.

Depending on the type of peripheral, half or full duplex, the flags belong to either one single channel or two different channels.

24.3.5.1 Receive Transfer End

This flag is set when PERIPH_RCR register reaches zero and the last data has been transferred to memory.

It is reset by writing a non zero value in PERIPH_RCR or PERIPH_RNCR.

24.3.5.2 Transmit Transfer End

This flag is set when PERIPH_TCR register reaches zero and the last data has been written into peripheral THR.

It is reset by writing a non zero value in PERIPH_TCR or PERIPH_TNCR.

24.3.5.3 Receive Buffer Full

This flag is set when PERIPH_RCR register reaches zero with PERIPH_RNCR also set to zero and the last data has been transferred to memory.

It is reset by writing a non zero value in PERIPH_TCR or PERIPH_TNCR.

24.3.5.4 Transmit Buffer Empty

This flag is set when PERIPH_TCR register reaches zero with PERIPH_TNCR also set to zero and the last data has been written into peripheral THR.

It is reset by writing a non zero value in PERIPH_TCR or PERIPH_TNCR.

24.4 Peripheral DMA Controller (PDC) User Interface

Table 24-1. Register Mapping

Offset	Register	Name	Access	Reset
0x100	Receive Pointer Register	PERIPH[(1)]_RPR	Read-write	0
0x104	Receive Counter Register	PERIPH_RCR	Read-write	0
0x108	Transmit Pointer Register	PERIPH_TPR	Read-write	0
0x10C	Transmit Counter Register	PERIPH_TCR	Read-write	0
0x110	Receive Next Pointer Register	PERIPH_RNPR	Read-write	0
0x114	Receive Next Counter Register	PERIPH_RNCR	Read-write	0
0x118	Transmit Next Pointer Register	PERIPH_TNPR	Read-write	0
0x11C	Transmit Next Counter Register	PERIPH_TNCR	Read-write	0
0x120	Transfer Control Register	PERIPH_PTCR	Write-only	0
0x124	Transfer Status Register	PERIPH_PTSR	Read-only	0

Note: 1. PERIPH: Ten registers are mapped in the peripheral memory space at the same offset. These can be defined by the user according to the function and the peripheral desired (DBGU, USART, SSC, SPI, MCI, etc.)

24.4.1 Receive Pointer Register

Register Name: PERIPH_RPR

Access Type: Read-write

31	30	29	28	27	26	25	24
RXPTR							
23	22	21	20	19	18	17	16
RXPTR							
15	14	13	12	11	10	9	8
RXPTR							
7	6	5	4	3	2	1	0
RXPTR							

- **RXPTR: Receive Pointer Register**

RXPTR must be set to receive buffer address.

When a half duplex peripheral is connected to the PDC, RXPTR = TXPTR.

24.4.2 Receive Counter Register

Register Name: PERIPH_RCR

Access Type: Read-write

31	30	29	28	27	26	25	24
–	–	–	–	–	–	–	–
23	22	21	20	19	18	17	16
–	–	–	–	–	–	–	–
15	14	13	12	11	10	9	8
RXCTR							
7	6	5	4	3	2	1	0
RXCTR							

- **RXCTR: Receive Counter Register**

RXCTR must be set to receive buffer size.

When a half duplex peripheral is connected to the PDC, RXCTR = TXCTR.

0 = Stops peripheral data transfer to the receiver

1 - 65535 = Starts peripheral data transfer if corresponding channel is active

24.4.3 Transmit Pointer Register

Register Name: PERIPH_TPR

Access Type: Read-write

31	30	29	28	27	26	25	24
TXPTR							
23	22	21	20	19	18	17	16
TXPTR							
15	14	13	12	11	10	9	8
TXPTR							
7	6	5	4	3	2	1	0
TXPTR							

- **TXPTR: Transmit Counter Register**

TXPTR must be set to transmit buffer address.

When a half duplex peripheral is connected to the PDC, RXPTR = TXPTR.

24.4.4 Transmit Counter Register

Register Name: PERIPH_TCR

Access Type: Read-write

31	30	29	28	27	26	25	24
–	–	–	–	–	–	–	–
23	22	21	20	19	18	17	16
–	–	–	–	–	–	–	–
15	14	13	12	11	10	9	8
TXCTR							
7	6	5	4	3	2	1	0
TXCTR							

- **TXCTR: Transmit Counter Register**

TXCTR must be set to transmit buffer size.

When a half duplex peripheral is connected to the PDC, RXCTR = TXCTR.

0 = Stops peripheral data transfer to the transmitter

1- 65535 = Starts peripheral data transfer if corresponding channel is active

24.4.5 Receive Next Pointer Register

Register Name: PERIPH_RNPR

Access Type: Read-write

31	30	29	28	27	26	25	24
RXNPTR							
23	22	21	20	19	18	17	16
RXNPTR							
15	14	13	12	11	10	9	8
RXNPTR							
7	6	5	4	3	2	1	0
RXNPTR							

- **RXNPTR: Receive Next Pointer**

RXNPTR contains next receive buffer address.

When a half duplex peripheral is connected to the PDC, RXNPTR = TXNPTR.

24.4.6 Receive Next Counter Register

Register Name: PERIPH_RNCR

Access Type: Read-write

31	30	29	28	27	26	25	24
–	–	–	–	–	–	–	–
23	22	21	20	19	18	17	16
–	–	–	–	–	–	–	–
15	14	13	12	11	10	9	8
RXNCTR							
7	6	5	4	3	2	1	0
RXNCTR							

- **RXNCTR: Receive Next Counter**

RXNCTR contains next receive buffer size.

When a half duplex peripheral is connected to the PDC, RXNCTR = TXNCTR.

24.4.7 Transmit Next Pointer Register

Register Name: PERIPH_TNPR

Access Type: Read-write

31	30	29	28	27	26	25	24
TXNPTR							
23	22	21	20	19	18	17	16
TXNPTR							
15	14	13	12	11	10	9	8
TXNPTR							
7	6	5	4	3	2	1	0
TXNPTR							

- **TXNPTR: Transmit Next Pointer**

TXNPTR contains next transmit buffer address.

When a half duplex peripheral is connected to the PDC, RXNPTR = TXNPTR.

24.4.8 Transmit Next Counter Register

Register Name: PERIPH_TNCR

Access Type: Read-write

31	30	29	28	27	26	25	24
–	–	–	–	–	–	–	–
23	22	21	20	19	18	17	16
–	–	–	–	–	–	–	–
15	14	13	12	11	10	9	8
TXNCTR							
7	6	5	4	3	2	1	0
TXNCTR							

- **TXNCTR: Transmit Counter Next**

TXNCTR contains next transmit buffer size.

When a half duplex peripheral is connected to the PDC, RXNCTR = TXNCTR.

24.4.9 Transfer Control Register

Register Name: PERIPH_PTCR

Access Type: Write-only

31	30	29	28	27	26	25	24
–	–	–	–	–	–	–	–
23	22	21	20	19	18	17	16
–	–	–	–	–	–	–	–
15	14	13	12	11	10	9	8
–	–	–	–	–	–	TXTDIS	TXTEN
7	6	5	4	3	2	1	0
–	–	–	–	–	–	RXTDIS	RXTEN

- **RXTEN: Receiver Transfer Enable**

0 = No effect.

1 = Enables PDC receiver channel requests if RXTDIS is not set.

When a half duplex peripheral is connected to the PDC, enabling the receiver channel requests automatically disables the transmitter channel requests. It is forbidden to set both TXTEN and RXTEN for a half duplex peripheral.

- **RXTDIS: Receiver Transfer Disable**

0 = No effect.

1 = Disables the PDC receiver channel requests.

When a half duplex peripheral is connected to the PDC, disabling the receiver channel requests also disables the transmitter channel requests.

- **TXTEN: Transmitter Transfer Enable**

0 = No effect.

1 = Enables the PDC transmitter channel requests.

When a half duplex peripheral is connected to the PDC, it enables the transmitter channel requests only if RXTEN is not set. It is forbidden to set both TXTEN and RXTEN for a half duplex peripheral.

- **TXTDIS: Transmitter Transfer Disable**

0 = No effect.

1 = Disables the PDC transmitter channel requests.

When a half duplex peripheral is connected to the PDC, disabling the transmitter channel requests disables the receiver channel requests.

24.4.10 Transfer Status Register

Register Name: PERIPH_PTSR

Access Type: Read-only

31	30	29	28	27	26	25	24
–	–	–	–	–	–	–	–
23	22	21	20	19	18	17	16
–	–	–	–	–	–	–	–
15	14	13	12	11	10	9	8
–	–	–	–	–	–	–	TXTEN
7	6	5	4	3	2	1	0
–	–	–	–	–	–	–	RXTEN

- **RXTEN: Receiver Transfer Enable**

0 = PDC Receiver channel requests are disabled.

1 = PDC Receiver channel requests are enabled.

- **TXTEN: Transmitter Transfer Enable**

0 = PDC Transmitter channel requests are disabled.

1 = PDC Transmitter channel requests are enabled.